3	**(a)**		$y > x+5$ $y \leq 8-2x-x^2$	**B1** **B1** **[2]**	**2.5** **2.5**	Allow interchange of $>$ and $\geq$ or $<$ and $\leq$ for one inequality as long as the direction is correct Both inequalities fully correct. Allow $x+5 < y \leq 8-2x-x^2$ oe	
3	**(b)**		**DR** Boundary values when $8-2x-x^2 = x+5$ $x^2+3x-3=0$ giving $x = \dfrac{-3\pm\sqrt{21}}{2}$ From the graph, the line lies below the curve for $\dfrac{-3-\sqrt{21}}{2} < x < \dfrac{-3+\sqrt{21}}{2}$ or $\left\{x : x > \dfrac{-3-\sqrt{21}}{2}\right\} \cap \left\{x : x < \dfrac{-3+\sqrt{21}}{2}\right\}$	**M1** **A1** **B1** **[3]**	**2.1** **2.1** **2.1**	A correct three term quadratic equation (or inequality) must be seen Correct roots of the equation soi Must be exact Shows correct inequality FT their roots	Any method including BC is acceptable for solving the quadratic equation clearly seen in the form $ax^2+bx+c=0$. Allow B1 for $-3.79 < x < 0.79$ www